

A system of perimeter cameras controlled wirelessly will create geofencing to keep specific assets in particular areas, warn ramp drivers about their speed, or immediately direct fuel trucks or other services to the relevant aircraft. It will be possible to integrate information technology in a modular manner, choosing a *la carte* from an extensive menu based on need and budget. Innovations in secure and reliable cloud technology will help substitute local infrastructure with shared service, making the technology more efficient and cost-effective, even for the smallest airports.

Enhanced passenger experience

Imagine boarding a flight after scanning your boarding pass, tagging and checking your bags and locating the nearest spot for an airport manicure by tapping a screen. Passengers will be able to weigh their own luggage without the assistance of an agent, and drop the bags off at luggage acceptance points. Do-it-yourself boarding will allow passengers to speed up the process of getting to their seats. Self-service kiosks will be available for passengers to resolve issues such as booking new flights if previous flight was cancelled.

Flight attendants will have a new, advanced type of mobile device that provides every bit of information right from the name and age of passengers who might need a little extra attention to notes about fliers' all previous flying experiences. They will carry phablets for communication with passengers, who will be able to text their requirement from their seat. The devices hand-



Flight attendants are carrying more advanced devices with passage of time (Image courtesy: www.technologyrecord.com)



Snapshot of a smart FIDS end-user screen (Image courtesy: www.travelwayfinding.com)

held by the crew could be fed with highly specific information, such as a flier's request not to wake him up for a meal. The devices could even be used to quicken maintenance—for example, by taking a picture of a damaged tray table and transmitting it to the mechanic at the other end to seek help in fixing right away.

Smart flight information display system

A smart flight information display system (FIDS) also includes the baggage information display system and advertising system. A key component of the airport's integrated operational systems, smart FIDS are directly connected to the airport's operational database and resource management. With the airport environments becoming increasingly intense, the flight and operational information needs to be disseminated quickly and cost-effectively. For this purpose, the FIDS include a

control centre, distribution servers, input terminals and display technology. The system control centre is responsible for storing, processing and transferring all flight and general information. The processed information is sent by FIDS via LAN to the connected distribution servers in real time so that the users are supplied with high-quality consistent information. This information is processed and transferred, by the distribution server, to the corresponding display devices.

The FIDS system communicates and supports any type of display device such as CRT monitors, LCD flat panels, plasma displays, split flap boards and LED signage. It is possible to

use new communication technologies to distribute and display the information in any required format, such as text, audio, video, images, animations and movies. The system is cost-effective and provides flexible state-of-the-art multimedia display. It is easily configurable and automatically collects the information, organises it according to programmed graphical and paging formats, and disseminates to keep passengers, operations staff, airport management, ground handling personnel or airlines crew informed. Information on the flight's status, boarding time, gate or baggage belt assignments, boarding gates, baggage sorting carousels, arrival and departure areas, etc is available in real time. It can play a key role in airport safety, for example, by providing immediate information for airport evacuation in case of an emergency or security alert. The fault tolerance option is